Mathematics of Machine Learning

Probability

$Y = g(X)$			Ω
$y_{13} = g(x_{13})$	$y_{14} = g(x_{14})$	$y_{15} = g(x_{15})$	$y_{16} = g(x_{16})$
$y_9 = g(x_9)$	$y_{10} = g(x_{10})$	$y_{11} = g(x_{11})$	$y_{12} = g(x_{12})$
$y_5 = g(x_5)$	$y_6 = g(x_6)$	$y_7 = g(x_7)$	$y_8 = g(x_8)$
$y_1 = g(x_1)$	$y_2 = g(x_2)$	$y_3 = g(x_3)$	$y_4 = g(x_4)$

Probability, Random Variables, Expectations

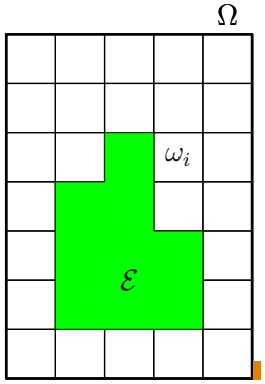
Outline

1. **Random Variables**
2. Expectations
3. Calculus of Probabilities

Ω				
x_{31}	x_{32}	x_{33}	x_{34}	x_{35}
x_{26}	x_{27}	x_{28}	x_{29}	x_{30}
x_{21}	x_{22}	x_{23}	x_{24}	x_{25}
x_{16}	x_{17}	x_{18}	x_{19}	x_{20}
x_{11}	x_{12}	x_{13}	x_{14}	x_{15}
x_6	x_7	x_8	x_9	x_{10}
x_1	x_2	x_3	x_4	x_5

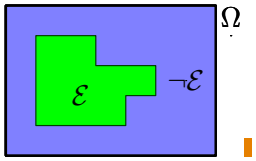
Modelling Uncertainty

- To model a world with uncertainty we consider some set of **elementary events** or **outcomes** Ω
- For the outcome of rolling a dice $\Omega = \{1,2,3,4,5,6\}$
- Outcomes are **mutually exclusive** (exactly one happens) and **exhaustive** (together they make up Ω)
- An **event** is a set of outcomes, $\mathcal{E} \subseteq \Omega$
- E.g. For a dice throw $\mathcal{E} = \{2,4,6\}$



Probabilities

- We attribute a **probability**, $\mathbb{P}(\mathcal{E})$, to an event, $\mathcal{E}$, with the requirements
  - $0 \leq \mathbb{P}(\mathcal{E}) \leq 1$, $\mathbb{P}(\Omega) = 1$
  - $\mathbb{P}(\mathcal{A} \cup \mathcal{B}) = \mathbb{P}(\mathcal{A}) + \mathbb{P}(\mathcal{B})$ if $\mathcal{A} \cap \mathcal{B} = \emptyset$
  - Hence $\mathbb{P}(\mathcal{E}) + \mathbb{P}(\neg\mathcal{E}) = 1$ where $\neg\mathcal{E} = \Omega \setminus \mathcal{E}$



- In some cases we can interpret $\mathbb{P}(\mathcal{E})$ as the expected frequency of occurrence of a repetitive trial
- But $\mathbb{P}(\text{Pass AICE3001 exam})$ is something you do once
- Think of probability as an informed belief, which changes when our knowledge changes

Probability Is Not Counting

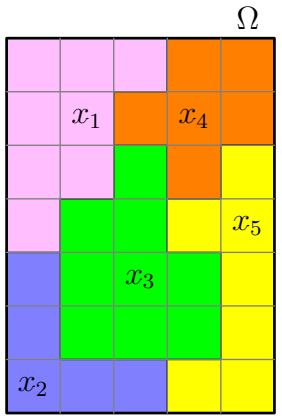
- At school probability is often taught as counting: if all outcomes are equally likely then $\mathbb{P}(\mathcal{E}) = |\mathcal{E}|/|\Omega|$
- But "equally likely" is an assumption about the world (a fair dice, a well-shuffled pack), not part of probability
- Most problems have no equally likely outcomes: tomorrow's weather, whether a network misclassifies an image
- Probability is about the **rules** for combining and updating beliefs: adding probabilities of exclusive events, conditioning, marginalising
- Combinatorics is just a tool for counting outcomes when symmetry tells us they are equally likely

Random Variables

- A **random variable**, X, assigns a number to each outcome in Ω: it partitions Ω and gives each part a number
- E.g. for a dice

$$X = \begin{cases} 0 & \text{if } \omega \in \{1,3,5\} \\ 1 & \text{if } \omega \in \{2,4,6\} \end{cases}$$

- $\mathbb{P}(X = x_i) = \mathbb{P}(\mathcal{E}_i)$ where $\mathcal{E}_i$ is the event corresponding to the part with value x_i



What's In A Name

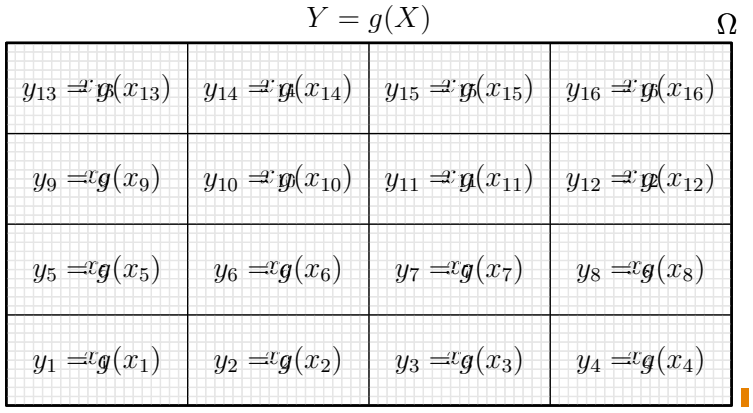
- We denote random variables with capital letters, X, Y, Z, etc.
- The symbol denotes an object that can take one of a number of different values, but which one is still to be decided by chance
- When we write $\mathbb{P}(X)$ we can view this as short-hand for

$$(\mathbb{P}(X = x) \mid x \in \mathcal{X}) = (\mathbb{P}(X = x_1), \mathbb{P}(X = x_2), \ldots \mathbb{P}(X = x_n))$$

  where $\mathcal{X}$ is the set of possible values that X can take
- We treat random variables very differently to normal numbers (scalars) when we consider taking expectations

Function of Random Variables

- Any function, $Y = g(X)$, of a random variable, X, is a random variable



Continuous Spaces

- If the space of elementary events is continuous (e.g. for darts $\boldsymbol{x} = (x,y)$) then $\mathbb{P}(\boldsymbol{X} = \boldsymbol{x}) = 0$
- But if we consider a region, $\mathcal{R}$, then we can assign a probability to landing in the region $\mathbb{P}(\boldsymbol{X} \in \mathcal{R})$
- It is useful to work with the **probability density function** (PDF)

$$f_{\boldsymbol{X}}(\boldsymbol{x}) = \lim_{\epsilon \to 0} \frac{\mathbb{P}(\boldsymbol{X} \in \mathcal{B}(\boldsymbol{x},\epsilon))}{\mathrm{Vol}(\mathcal{B}(\boldsymbol{x},\epsilon))}$$



where $\mathcal{B}(\boldsymbol{x},\epsilon)$ is a ball of radius ϵ around the point $\boldsymbol{x}$ and $\mathrm{Vol}(\mathcal{B}(\boldsymbol{x},\epsilon))$ is the volume of the ball

- If we make a change of variable the volume $\mathrm{Vol}(\mathcal{B}(\boldsymbol{x},\epsilon))$ might change so $f_{\boldsymbol{X}}(\boldsymbol{x})$ will change

Change of Variables

- Consider a region $\mathcal{R}$—we can describe this using different coordinate systems x or $y = g(x)$
- But

$$\mathbb{P}(X \in \mathcal{R}) = \int_{\mathcal{R}} f_X(x)\mathrm{d}x = \mathbb{P}(Y \in \mathcal{R}) = \int_{\mathcal{R}} f_Y(y)\mathrm{d}y$$

- As this is true for any region $\mathcal{R}$: $f_X(x)|\mathrm{d}x| = f_Y(y)|\mathrm{d}y|$
- Or

$$f_X(x) = f_Y(y)\left|\frac{\mathrm{d}y}{\mathrm{d}x}\right| = f_Y(g(x))|g'(x)|$$

- A probability density measured in units of probability per cm is different to that measured in units of probability per inch

Jacobian

- In high dimension if we make a change of variables $\boldsymbol{x} \to \boldsymbol{y}(\boldsymbol{x})$ (which can be seen as a change of random variables $\boldsymbol{X} \to \boldsymbol{Y}(\boldsymbol{X})$)
- Then

$$f_{\boldsymbol{X}}(\boldsymbol{x}) = f_{\boldsymbol{Y}}(\boldsymbol{y})|\det(\mathbf{J})|$$

where $\mathbf{J}$ is the Jacobian matrix

$$\mathbf{J} = \begin{pmatrix} \frac{\partial y_1}{\partial x_1} & \frac{\partial y_1}{\partial x_2} & \cdots & \frac{\partial y_1}{\partial x_n} \\ \frac{\partial y_2}{\partial x_1} & \frac{\partial y_2}{\partial x_2} & \cdots & \frac{\partial y_2}{\partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial y_n}{\partial x_1} & \frac{\partial y_n}{\partial x_2} & \cdots & \frac{\partial y_n}{\partial x_n} \end{pmatrix}$$

- Ensures integrals over volumes are the same

Meaning of Probability Densities

- Probability densities are not probabilities
- They are non-negative, but can be greater than 1
- Note that

$$f_X(x) = \lim_{\delta x \to 0} \frac{\mathbb{P}(x \le X < x + \delta x)}{\delta x}$$

- We can think of $f_X(x)\delta x$ as $\mathbb{P}(x \le X < x + \delta x)$
- Note that $f_X(x)\delta x \le 1$

Cumulative Distribution Functions

- We can define the **cumulative distribution function** (CDF)

$$F_X(x) = \mathbb{P}(X \leq x) = \begin{cases} \sum_{i:x_i \leq x} \mathbb{P}(X = x_i) \\ \int_{-\infty}^{x} f_X(y)\mathrm{d}y \end{cases}$$

- This is a function that goes from 0 to 1 as x goes from $-\infty$ to ∞
- We note that for continuous random variables

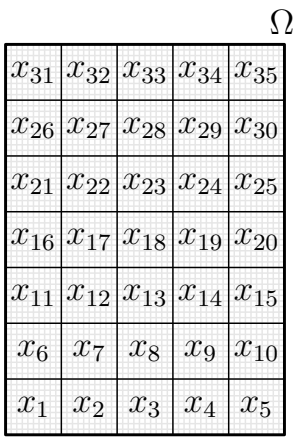
$$f_X(x) = \frac{\mathrm{d}F_X(x)}{\mathrm{d}x}$$

The Language of Probability

- Mathematical texts use a standard vocabulary
  - ⋆ **sample space** Ω, **outcome** ω, **event** $\mathcal{E} \subseteq \Omega$
  - ⋆ **random variable** X: a function from outcomes to numbers
  - ⋆ its **distribution**: a **probability mass function** $\mathbb{P}(X = x)$ (discrete) or **density** $f_X(x)$ (continuous), or the CDF $F_X(x)$
  - ⋆ $X \sim \mathcal{N}(\mu,\sigma^2)$: "$X$ is distributed as" a normal
  - ⋆ **support**: the values with non-zero probability (or density)
  - ⋆ **i.i.d.**: independent and identically distributed (e.g. repeated, well conducted experiments)
  - ⋆ **almost surely**: with probability one

Outline

1. Random Variables
2. **Expectations**
3. Calculus of Probabilities



Expectation

- We can define the expectation of $\boldsymbol{Y} = g(\boldsymbol{X})$ as

$$\mathbb{E}_{\boldsymbol{X}}[g(\boldsymbol{X})] = \begin{cases} \sum_{\boldsymbol{x}\in\mathcal{X}} g(\boldsymbol{x})\mathbb{P}(\boldsymbol{X} = \boldsymbol{x}) \\ \int g(\boldsymbol{x}) f_{\boldsymbol{X}}(\boldsymbol{x})\mathrm{d}\boldsymbol{x} \end{cases}$$

- The expectation of a constant c is

$$\mathbb{E}_{\boldsymbol{X}}[c] = \begin{cases} \sum_{\boldsymbol{x}\in\mathcal{X}} c\mathbb{P}(\boldsymbol{X} = \boldsymbol{x}) = c\sum_{\boldsymbol{x}\in\mathcal{X}} \mathbb{P}(\boldsymbol{X} = \boldsymbol{x}) = c \\ \int c f_{\boldsymbol{X}}(\boldsymbol{x})\mathrm{d}\boldsymbol{x} = c\int f_{\boldsymbol{X}}(\boldsymbol{x})\mathrm{d}\boldsymbol{x} = c \end{cases}$$

- Note $\mathbb{E}_X[\mathbb{E}_X[g(X)]] = \mathbb{E}_X[g(X)]$

Linearity of Expectation

- Because sums and integrals are linear operators

$$\sum_i (ax_i + by_i) = a\left(\sum_i x_i\right) + b\left(\sum_i y_i\right)$$

$$\int (af(\boldsymbol{x}) + bg(\boldsymbol{x}))\mathrm{d}\boldsymbol{x} = a\left(\int f(\boldsymbol{x})\mathrm{d}\boldsymbol{x}\right) + b\left(\int g(\boldsymbol{x})\mathrm{d}\boldsymbol{x}\right)$$

  then expectations are linear

$$\mathbb{E}[aX + bY] = a\mathbb{E}[X] + b\mathbb{E}[Y]$$

- Beware: in general $\mathbb{E}[XY] \neq \mathbb{E}[X]\mathbb{E}[Y]$ (equality holds if X and Y are independent, or more generally uncorrelated)

Indicator Functions

- An indicator function has the property

$$[\![predicate]\!] = \begin{cases} 1 & \text{if } predicate \text{ is True} \\ 0 & \text{if } predicate \text{ is False} \end{cases}$$

  (sometimes written $\boldsymbol{I}_A(x)$ where $A(x)$ is the predicate)

- We can obtain probabilities from expectations

$$\mathbb{P}(predicate) = \mathbb{E}[[\![predicate]\!]]$$

- E.g. The CDF is given by

$$F_X(x) = \mathbb{P}(X \leq x) = \mathbb{E}[[\![X \leq x]\!]]$$

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x_6	x_7	x_8	x_9	x_{10}
x_1	x_2	x_3	x_4	x_5

Joint Probabilities

- Often we want to model complex processes where we have multiple random variables

- We can define the joint probability

$$p_{X,Y}(x,y) = \mathbb{P}(X = x, Y = y)$$

  i.e. the probability of the event where both $X = x$ and $Y = y$

- Clearly $\mathbb{P}(X,Y) = \mathbb{P}(Y,X)$

Marginalisation

- Probabilities are extremely easy to manipulate (although lots of people struggle)
- One of the most useful properties is known as **marginalisation**

$$\mathbb{P}(X) = \sum_{y \in \mathcal{Y}} \mathbb{P}(X, Y = y)$$

  where $\mathcal{Y}$ is the set of values that the random variable Y takes
- Note that when we write $\mathbb{P}(X)$ we are saying this is true for all values that X can take
- Although obvious and easy this is extremely useful

Conditional Probability

- We can also define the probability of an event X given that $Y = y$ has occurred

$$\mathbb{P}(X \mid Y = y) = \frac{\mathbb{P}(X, Y = y)}{\mathbb{P}(Y = y)}$$

  (for $\mathbb{P}(Y = y) > 0$)
- In constructing a model it is often much easier to specify conditional probabilities (because you know something) rather than joint probabilities
- When manipulating probabilities it is often easier to work with joint probabilities because we can simplify them by marginalising out random variables we are not interested in

Basic Calculus

- To obtain the joint probability we can use

$$\mathbb{P}(X, Y) = \mathbb{P}(X|Y)\mathbb{P}(Y) = \mathbb{P}(Y|X)\mathbb{P}(X)$$

- This generalises to more random variables

$$\mathbb{P}(X, Y, Z) = \mathbb{P}(X, Y|Z)\mathbb{P}(Z) = \mathbb{P}(X|Y, Z)\mathbb{P}(Y|Z)\mathbb{P}(Z)$$

- We can do this in a number of different ways

$$\mathbb{P}(X, Y, Z) = \mathbb{P}(Y, Z|X)\mathbb{P}(X) = \mathbb{P}(Z|Y, X)\mathbb{P}(Y|X)\mathbb{P}(X)$$

- Note that $\mathbb{P}(A, B \mid X, Y)$ means the probability of random variables A and B given that X and Y take particular values

Beware

- Conditional probabilities, $\mathbb{P}(X \mid Y)$ are probabilities for X, but not Y

$$\sum_{x \in \mathcal{X}} \mathbb{P}(X = x \mid Y) = 1$$

$$\sum_{y \in \mathcal{Y}} \mathbb{P}(X \mid Y = y) \neq 1$$

  (in general)
- Note that

$$\mathbb{E}_Y[\mathbb{P}(X \mid Y)] = \sum_{y \in \mathcal{Y}} \mathbb{P}(Y = y)\mathbb{P}(X|Y = y)$$

$$= \sum_{y \in \mathcal{Y}} \mathbb{P}(X, Y = y) = \mathbb{P}(X)$$

Causality

- Conditional probabilities do not imply causality
- We might have causal relationships

$$\mathbb{P}(\text{pass} \mid \text{study}) = 0.9 \qquad \mathbb{P}(\text{pass} \mid \neg\text{study}) = 0.2$$

- But if we know $\mathbb{P}(\text{study}) = 0.8$ then we can compute

$$\mathbb{P}(\text{pass},\text{study}) = \mathbb{P}(\text{pass} \mid \text{study})\mathbb{P}(\text{study}) = 0.9 \times 0.8 = 0.72$$
$$\mathbb{P}(\text{pass},\neg\text{study}) = \mathbb{P}(\text{pass} \mid \neg\text{study})\mathbb{P}(\neg\text{study}) = 0.2 \times 0.2 = 0.04$$

and

$$\mathbb{P}(\text{study} \mid \text{pass}) = \frac{\mathbb{P}(\text{pass},\text{study})}{\mathbb{P}(\text{pass})} = \frac{\mathbb{P}(\text{pass},\text{study})}{\mathbb{P}(\text{pass},\text{study}) + \mathbb{P}(\text{pass},\neg\text{study})} = \frac{0.72}{0.72 + 0.04} \approx 0.947$$

Independence

- Random variables X and Y are said to be **independent** if

$$\mathbb{P}(X,Y) = \mathbb{P}(X)\mathbb{P}(Y)$$

- Because $\mathbb{P}(X,Y) = \mathbb{P}(X|Y)\mathbb{P}(Y)$ and $\mathbb{P}(X,Y) = \mathbb{P}(Y|X)\mathbb{P}(X)$ independence implies

$$\mathbb{P}(X|Y) = \mathbb{P}(X) \qquad \mathbb{P}(Y|X) = \mathbb{P}(Y)$$

- Probabilistic independence is a mathematical property, not necessarily causal independence
- Variables with no causal link can be dependent through a shared cause (ice-cream sales and drownings)
- If $X \in \{0,1\}$ represents the outcome of tossing a coin and $Y \in \{1,2,3,4,5,6\}$ the outcome of rolling a dice then X and Y are independent

Well Conducted Experiments

- In well conducted experiments we expect the results we obtain are independent
- Let $\mathcal{D} = (X_1,X_2,\ldots,X_m)$ represents possible outcomes from a set of m well conducted experiments then

$$\mathbb{P}(\mathcal{D}) = \prod_{i=1}^{m} \mathbb{P}(X_i)$$

- Denoting a possible sentence I might say by $\mathcal{S} = (W_1,W_2,\ldots,W_m)$ then

$$\mathbb{P}(\mathcal{S}) \neq \prod_{i=1}^{m} \mathbb{P}(W_i)$$

otherwise it's time I retired

Conditional Independence

- Let $K(d)$ be a random variable measuring the amount you know about ML on day d of your revision
- From your revision schedule you can write down your belief

$$\mathbb{P}(K(d) \mid K(d-1),K(d-2),\ldots K(1))$$

- But a very reasonable model is

$$\mathbb{P}(K(d) \mid K(d-1),K(d-2),\ldots K(1)) = \mathbb{P}(K(d) \mid K(d-1))$$

what you are going to know today will just depend on what you knew yesterday

- We say that $K(d)$ is **conditionally independent** of $K(d-2)$, $K(d-3)$, etc. given $K(d-1)$

Conclusion

- To work with probabilities you need to know
  - How to go back and forward between joint probabilities and conditional probabilities
  - How to marginalise out variables
- You need to understand that for continuous outcomes, it makes sense to talk about the probability density
- You need to know that expectations are linear operators and the expectation of a constant is the constant
- You need to understand independence